

Estimated persuasive effect (pp, 95% CI)

12
8
4
0

Static message
(6.66pp, GPT-4.5, Study 3)

1e+01

1e+02

1e+03

1e+04

1e+05

Effective compute (FLOPs 1e21)

Model: Q = qwen1.5 L = llama3.1

Post-training: ● Chat ● Developer

Study: ● 1 ▲ 2 ■ 3

Q-0.5b

Q-1.8b

Q-4b

Q-7b

Q-14b

Q-32b

Q-72b-chat

Q-110b-chat

Q-140b

Q-175b

Q-220b

Q-260b

Q-300b

Q-350b

Q-400b

Q-450b

Q-500b

Q-550b

Q-600b

Q-650b

Q-700b

Q-750b

Q-800b

Q-850b

Q-900b

Q-950b

Q-1000b

Q-1050b

Q-1100b

Q-1150b

Q-1200b

Q-1250b

Q-1300b

Q-1350b

Q-1400b

Q-1450b

Q-1500b

Q-1550b

Q-1600b

Q-1650b

Q-1700b

Q-1750b

Q-1800b

Q-1850b

Q-1900b

Q-1950b

Q-2000b

Q-2050b

Q-2100b

Q-2150b

Q-2200b

Q-2250b

Q-2300b

Q-2350b

Q-2400b

Q-2450b

Q-2500b

Q-2550b

Q-2600b

Q-2650b

Q-2700b

Q-2750b

Q-2800b

Q-2850b

Q-2900b

Q-2950b

Q-3000b

Q-3050b

Q-3100b

Q-3150b

Q-3200b

Q-3250b

Q-3300b

Q-3350b

Q-3400b

Q-3450b

Q-3500b

Q-3550b

Q-3600b

Q-3650b

Q-3700b

Q-3750b

Q-3800b

Q-3850b

Q-3900b

Q-3950b

Q-4000b

Q-4050b

Q-4100b

Q-4150b

Q-4200b

Q-4250b

Q-4300b

Q-4350b

Q-4400b

Q-4450b

Q-4500b

Q-4550b

Q-4600b

Q-4650b

Q-4700b

Q-4750b

Q-4800b

Q-4850b

Q-4900b

Q-4950b

Q-5000b

Q-5050b

Q-5100b

Q-5150b

Q-5200b

Q-5250b

Q-5300b

Q-5350b

Q-5400b

Q-5450b

Q-5500b

Q-5550b

Q-5600b

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Q-5900b

Q-5950b

Q-6000b

Q-6050b

Q-6100b

Q-6150b

Q-6200b

Q-6250b

Q-6300b

Q-6350b

Q-6400b

Q-6450b

Q-6500b

Q-6550b

Q-6600b

Q-6650b

Q-6700b

Q-6750b

Q-6800b

Q-6850b

Q-6900b

Q-6950b

Q-7000b

Q-7050b

Q-7100b

Q-7150b

Q-7200b

Q-7250b

Q-7300b

Q-7350b

Q-7400b

Q-7450b

Q-7500b

Q-7550b

Q-7600b

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Q-8000b

Q-8050b

Q-8100b

Q-8150b

Q-8200b

Q-8250b

Q-8300b

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Q-10000b

Q-10050b

Q-10100b

Q-10150b

Q-10200b

Q-10250b

Q-10300b

Q-10350b

Q-10400b

Q-10450b

Q-10500b

Q-10550b

Q-10600b

Q-10650b

Q-10700b

Q-10750b

Q-10800b

Q-10850b

Q-10900b

Q-10950b

Q-11000b

Q-11050b

Q-11100b

Q-11150b

Q-11200b

Q-11250b

Q-11300b

Q-11350b

Q-11400b

Q-11450b

Q-11500b

Q-11550b

Q-11600b

Q-11650b

Q-11700b

Q-11750b

Q-11800b

Q-11850b

Q-11900b

Q-11950b

Q-12000b

Q-12050b

Q-12100b

Q-12150b

Q-12200b

Q-12250b

Q-12300b

Q-12350b

Q-12400b

Q-12450b

Q-12500b

Q-12550b

Q-12600b

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Q-14100b

Q-14150b

Q-14200b

Q-14250b

Q-14300b

Q-14350b

Q-14400b

Q-14450b

Q-14500b

Q-14550b

Q-14600b

Q-14650b

Q-14700b

Q-14750b

Q-14800b

Q-14850b

Q-14900b

Q-14950b

Q-15000b

Q-15050b

Q-15100b

Q-15150b